

1. Pattern Recognition Systems

A. List and describe the four core components of a supervised pattern recognition system.

- I. Sensor: Collects raw data from the environment.
- II. Feature extraction: Transforms raw data into a set of measurable and relevant features that can be used for classification.
- III. Classifier: Uses the extracted features to assign the input to one of the predefined classes.
- IV. Training Component : It's made up by learning algorithm and teacher. It learns the mapping between features and labels using a training dataset with known labels.

B. Explain how the roles of the “teacher” and “learning algorithm” interact in training.

In supervised learning the teacher is the dataset that provides labeled examples. The learning algorithm uses these labeled examples to adjust the internal parameters of the classifier. It minimizes classification error or loss function between the predicted output and the true labels provided by the teacher. So the teacher provides correct outputs for given inputs and the learning algorithm uses these to compute a loss and adjust the model to better predict those labels in the future.

C. Illustrate the difference between training mode and classification mode.

- I. Training Mode: It learns a model from labeled data. The inputs are features and known labels. The output is a trained classifier. It involves feature extraction and parameter optimization.
- II. Classification Mode: It uses the model to assign class labels to new data. The inputs are only the features. The output is the predicted label. It involves feature extraction and applying decision rule.

2. Classification vs. Regression

A. Define the main difference between classification and regression.

In classification the goal is to assign input data to one of several discrete classes and the output is categorical. But in regression the goal is to predict a continuous numerical value based on the input data and the output is continuous.

B. For each, identify the type of output variable and give one application from the slides.

Output variable type for classification is categorical which could be nominal or ordinal and for regression is continuous. Classification can be applied for e-mail spam detection. It classifies as spam or not spam. Regression can be applied for a pricing prediction for example house price prediction.

3. Feature Vectors and Hidden States

A. Define a feature vector and explain what it represents in pattern recognition.

A feature vector is a numerical representation of an object or observation used in pattern recognition and machine learning. It is an ordered vector of measurable features extracted from raw data. It encodes all relevant information needed by a classifier to make a decision about the input. It transforms complex data for example images into a fixed-size numerical format that can be used by a classification or regression algorithm.

B. What is a hidden state and how is it related to the classification task?

A hidden state refers to an internal or latent variable that influences the observed data but is not directly observable. Hidden states represent underlying conditions or categories that generate the observable features. In many classification tasks the true class of an input for example the emotional state is a hidden state. The classification task involves inferring this hidden state from the observed feature vector.

4. Image Based: Template Matching

A. Based on the image below, explain how template matching classifies input patterns using training templates.

Template Matching is a nearest-neighbor method in which classification is based on comparing an input pattern to stored templates which are examples from the training set. Each input image is represented as a feature vector by flattening the 12×12 binary matrix into a 144-dimensional vector of values of 0 or 1. The system then stores templates of each class. For example in letter recognition it would be templates of A, B, When a new input is received the system computes a similarity or distance between the input vector and each stored template. Then the input is classified as the class of the most similar template

B. Why does template matching struggle when fewer templates are stored?

If fewer templates are stored some images may not be recognized. So with fewer templates the classifier lacks the representation power to recognize all acceptable forms of a given class, leading to higher misclassification rates.

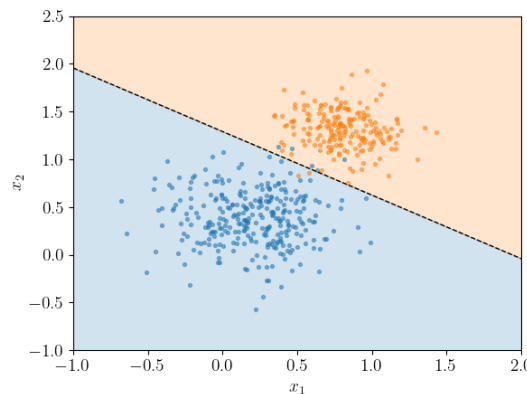
5. Decision Regions and Boundaries

A. What is a decision region in classification?

A decision region is a subset of the input feature space in which all points are assigned to the same class by a classifier. The input space is partitioned into regions, each corresponding to a class label. Every input within a region will be classified the same way.

B. Define and illustrate the concept of a decision boundary.

A decision boundary is the line that separates decision regions of different classes. It represents the transition points in the feature space where the classifier switches from one class label to another. The image below shows the decision boundary of a logistic regression model.



C. How does uncertainty affect the clarity of decision regions?

Uncertainty blurs the boundaries between decision regions, making them less distinct. Classifiers may give low confidence predictions, and decision boundaries may zigzag or be unstable. This could be due to overlapping data distributions, noisy inputs or insufficient training data.

6. Bayes Classification Rule

A. State Bayes' theorem in the context of classification.

Bayes' theorem relates the probability of a class given some observed features (posterior) to the likelihood of observing those features under that class.

$$P(w_i | x) = \frac{p(x | w_i)P(w_i)}{p(x)}$$

B. Derive the decision rule for assigning an observation to a class.

Because we want to assign a feature vector x to the most probable class given the observation so we use the Bayes Decision Rule and assign x to the class w_i that maximizes $P(w_i|x)$. So the

decision rule becomes the following. The following formula means classify x as the class with the highest product of likelihood and prior.

$$C = \arg \max [P(x|w_i) \cdot P(w_i)]$$

C. Explain the terms: prior, likelihood, posterior, and evidence.

- I. Prior $P(w_i)$: The probability of class w_i before observing data.
 - II. Likelihood $P(x|w_i)$: The probability of observing the feature vector x if the true class is w_i .
 - III. Posterior $P(w_i|x)$: The probability that the input belongs to class w_i given the data.
 - IV. Evidence $p(x)$: The total probability of observing x under all possible classes.
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7. Probability Density Function Estimation

A. Why is estimating the PDF of feature vectors important for Bayes classifiers?

PDF estimation is critical because Bayes classifiers rely on it to model how likely it is to observe a given input under each class.

B. Compare parametric vs. non-parametric estimation, with one example each.

- I. For parametric estimation we imagine that the features for each class follow a multivariate Gaussian distribution. First we estimate the mean and covariance matrix for each class using training data then we use the Gaussian PDF formula for $p(x|w_i)$. Parametric estimation assumes a specific functional form. It has fixed number of parameters. It is typically faster and more efficient but can be less flexible.
 - II. For non-parametric estimation we can use KDE. In KDE each data point contributes a small bump like a Gaussian kernel and the PDF is the sum of all bumps. Non-parametric estimation has no assumptions about the distribution. The model complexity grows with the amount of data. It is slower especially if we have large datasets but it is very flexible.
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8. Image-Based: Decision Boundaries

A. Observe the figure below showing decision regions. Describe how the classifier forms the boundaries.

- I. The image on the right has complex boundaries and they are overfit to the training data. It could be result of Parzen Window
- II. The image on the right has smoother decision boundaries and it could be result of K-NN. This classifier generalizes better.

B. What happens to the boundaries if classes have overlapping features?

Then they blur the decision boundaries and then boundaries shift toward regions of lower density. There is a trade-off between precision and generality.

9. 1-NN and Similarity-Based Methods

- A. Explain how the 1-Nearest Neighbor rule works using similarity.

Given a new input x , the algorithm computes the distance between x and all training examples. Then finds the closest training sample and assigns the class of that nearest neighbor to x . 1-NN assumes that similar points have similar labels and it uses distance as a proxy for inverse similarity so this means the smaller the distance the more similar the points are.

- B. Why is the 1-NN considered implementation-independent?

Because it makes no assumptions about the underlying distribution of the data. It doesn't require a training phase or model-fitting process. It can be used with any data type like numeric, categorical and mixed as long as a suitable distance metric is defined.

- C. How does the choice of metric affect classifier accuracy?

It affects classification accuracy. A poor choice of metric can cause the algorithm to consider irrelevant or misleading features as close. The distance metric defines similarity so it affects both accuracy and boundary quality.

10. Parzen and k-NN Density Estimation

- A. Describe the Parzen window method for estimating a PDF.

The Parzen window method is a non-parametric way to estimate the PDF of a dataset. This method places a window which is a kernel function at the center of each training point. The kernel which is often Gaussian or uniform defines the contribution of each point to the estimated density. The PDF at any point x is estimated by summing all kernel contributions over the dataset.

$$\hat{p}(x) = \frac{1}{N} \sum_{i=1}^N \left(\frac{N_i}{h} \varphi \left(\frac{c_i - x}{h} \right) \right)$$

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- B. How does k-NN PDF estimation differ from Parzen?

K-NN is also non-parametric but adaptive. Parzen uses a fixed smoothing scale but k-NN uses a fixed number of neighbors but adapts the region based on local data density.

- C. What are the challenges in selecting parameters h or k ?

Selecting h or k requires balancing bias and variance.

- I. If h is too small estimation is noisy and has a high variance. This results in overfitting. If h is too large the estimation is too smooth and has a high bias. This results in underfitting. So we need careful tuning and we usually use cross-validation.
- II. If k is too small PDF is too spiky and it's sensitive to noise. If k is too large PDF is over-smoothed.

11. Discriminant Functions

A. What is a discriminant function?

A discriminant function is a function that assigns a numerical score to an input feature vector x , used to decide which class it belongs to. For a set of classes $\omega_1, \omega_2, \dots, \omega_C$ we define one discriminant function per class: $g_i(x)$ for class ω_i . The classification rule is to assign x to the class ω_i for which $g_i(x)$ is maximum.

B. Write the discriminant function for multivariate Gaussian classes.

$$g_i(x) = -\frac{1}{2} (x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i) - \frac{1}{2} \log |\Sigma_i| + \log P(\omega_i)$$

- μ_i : mean vector of class ω_i
- Σ_i : covariance matrix of class ω_i
- $P(\omega_i)$: prior probability of class ω_i
- $|\Sigma_i|$: determinant of the covariance matrix

C. How does this function relate to decision boundaries?

The decision boundary between two classes ω_i and ω_j is the set of points where $g_i(x) = g_j(x)$. This is the region where the classifier is indifferent between the two classes.

12. Ensemble Learning and EM algorithm

A. Explain how the EM algorithm is used to estimate parameters in a Gaussian mixture.

The EM algorithm is used to estimate the parameters of a GMM when the data is unlabeled. The EM algorithm iteratively estimates GMM parameters using the E-step or expectation which assigns soft labels and then the M-step or maximization which updates parameters. It maximizes the data likelihood.

B. Why might ensemble methods outperform a single classifier?

Combining multiple weak learners reduces the variance and bias depending on the method. Ensemble methods are also more robust to overfitting and noisy data.